

And the p.d.f of  $s'^2$  is,

$$u(s'^2) = \frac{(n-1)^{(n-1)/2}}{(2\sigma^2)^{(n-1)/2} \Gamma\left(\frac{n-1}{2}\right)} \cdot \exp[-n-1)s'^2/2\sigma^2](s'^2)^{n-3/2}, 0 < s'^2 < \infty$$

## 29.5 CENTRAL LIMIT THEOREM

For any positive constant  $c$ , the probability that  $\bar{X}$  will take a value  $(\mu - c)$  and  $(\mu + c)$  is at least  $1 - \frac{\sigma^2}{nc^2}$ . When  $n \rightarrow \infty$ , the probability tends towards 1. The result is called the law of large numbers and is of critical interest in theoretical statistics. One of its important applications is the Central Limit theorem, which concerns the limiting distribution of the standardized mean of  $n$  random variable when  $n \rightarrow \infty$ .

The central limit theorem may be stated as follows:

If  $x_i (i = 1, 2, \dots, n)$  are independently distributed random variables such that  $E(x_i) = \mu_i$  and  $V(x_i) = \sigma_i^2$ , then under certain very general conditions, the random variables  $S_n = x_1 + x_2 + \dots + x_n$ , is asymptotically normal with mean  $\mu$  and standard deviation  $\sigma$  where

$$\mu = \sum_{i=1}^n \mu_i \text{ and } \sigma^2 = \sum_{i=1}^n \mu_i^2$$

This result helps us in dealing with the observations which are not randomly distributed.

### Check Your Progress 2

- Given a test that is normally distributed with mean=30 and a standard deviation = 6, what is the probability that a single score drawn at random will be greater than 34?

.....  
 .....  
 .....  
 .....

- Assume a normal distribution with a mean of 90 and a standard deviation of 7. What limits would include the middle 65% of the cases?

.....  
 .....  
 .....

## 29.6 LET US SUM UP

This unit presented different types of sampling distributions. We learnt about the concept of critical value or significant value of different distributions, their probability density function and Jacobian transformation. Finally, we also had the intuitive idea about central limit theorem, which is useful for the non-randomly distributed sample.

## 29.7 KEY WORDS

**Central Limit Theorem** : If  $x_i$  ( $i = 1, 2, \dots, n$ ) are independently distributed random variables such that  $E(x_i) = \mu_i$  and  $V(x_i) = \sigma_i^2$  then, under certain assumption,  $S_n \left( = \sum_{i=1}^n x_i \right)$  is asymptotically normal with mean  $\mu$  and variance  $\sigma^2$  where

$$\mu = \sum_{i=1}^n \mu_i, \sigma^2 = \sum_{i=1}^n \sigma_i^2.$$

**Chi-square Distribution** : The square of a standard normal variate is known as chi-square variate with 1 df.

**F Distribution** : If  $X \sim x_{n_1}^2$  and  $Y \sim x_{n_2}^2$  and if  $X$  and  $Y$  are independent to each other then F statistic is defined by

$$F = \frac{X/n_1}{Y/n_2} \sim F_{(n_1, n_2)}$$

**Fisher's 't' Distribution** : If  $t \sim N(0,1)$  and  $x^2 \sim x_n^2$ , and if both are independent, then Fisher's 't' is given by  $t = t / \sqrt{x^2/n} \sim t_n$

**Standard Normal Distribution** : A continuous random variable, which is normally distributed with mean zero, variance 1 is called standard normal distribution.

**Student's 't' Distribution** : If  $x_i$  ( $i = 1, 2, \dots, n$ ) is a random sample of size  $n$  from a normal population with mean  $\mu$  and variance  $\sigma^2$ , then student's 't' is defined by the statistic  $t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim t_{(n-1)}.$

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## 29.8 ANSWER OR HINTS TO CHECK YOUR PROGRESS EXERCISES

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### Check Your Progress 1

- 1) See Section 29.3
- 2)  $\chi^2 = 14.5$  df = 5, p = .017
- 3) See Section 29.3
- 4) Hint: Here we set up the null hypothesis that the digit occurs equally frequently in the directory. Under the null hypothesis, the expected frequency for each of the digits 0,1,2,...,9 is  $10000/10=1000$ .

The value of  $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 58.542$

Here  $E_i = 1000 \forall i = 0,1,\dots,9$

$O_i$  = observed frequency given in the table.

Degree of freedom =  $10 - 1 = 9$  (since we are given the total frequencies, we are subjected to only one linear constraint  $\sum_i O_i = \sum_i E_i = 10000$ ).

The tabulated  $\chi_{0.05,9}^2 = 16.919 < 58.542$

Thus, we conclude that digits are not uniformly distributed.

### Check Your Progress 2

- 1) 0.2524
- 2) 83.46 and 96.54

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## 29.9 EXERCISES

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- Q1.** The theory predicts the proportion of bins in the four groups A, B, C and D should be 9:3:3:1. In an experiment among 1600 bins, the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory?

**Ans.** Total no. of bins = 1600

$$E(882) = \frac{9}{16} \times 1600 = 900$$

$$E(313) = \frac{3}{16} \times 1600 = 300 \text{ and so on}$$

$$\chi^2 = \sum \left[ \frac{(O - E)^2}{E} \right] = 4.7266 < \chi_{0.05,3}^2 (= 7.815)$$

$\therefore$  The null hypothesis is accepted.]

**Q2.** The CEO of light bulbs manufacturing company claims that an average light bulb lasts 300 days. A researcher randomly selects 15 bulbs for testing. The sampled bulbs last an average of 290 days, with a standard deviation of 50 days. If the CEO's claim were true, what is the probability that 15 randomly selected bulbs would have an average life of no more than 290 days?

**Ans:** The traditional approach requires you to compute the t statistic, based on data presented in the problem description.

The first thing we need to do is compute the t statistic, based on the following equation :  $t = \frac{\bar{x} - \mu}{(s/\sqrt{n})}$  where  $\bar{x}$  is the sample mean,  $\mu$  is the population mean,  $s$  is the standard deviation of the sample, and  $n$  is the sample size.

Therefore,  $t = (290 - 300) / (50 / \sqrt{15})$

$= -10 / 12.909945$

$= -0.7745966$

- The degrees of freedom are equal to  $15 - 1 = 14$ .
- Assuming the CEO's claim is true, the population mean equals 300.
- The sample mean equals 290.
- The standard deviation of the sample is 50.

The cumulative probability is 0.226. Hence, if the true bulb life were 300 days, there is a 22.6% chance that the average bulb life for 15 randomly selected bulbs would be less than or equal to 290 days.

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## UNIT 30 ESTIMATION

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### Structure

- 30.0 Objectives
- 30.1 Introduction
- 30.2 Theory of Estimation
  - 30.2.1 Parameter Space
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  - 30.3.2 Consistency
  - 30.3.3 Efficiency
    - 30.3.3.1 Most Efficient Estimator
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- 30.4 Cramer Rao Inequality
  - 30.4.1 MVUE and Blackwellisation
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- 30.5 Asymptotic Properties
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- 30.6 Methods of Estimation
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  - 30.6.3 Partial Least Squares
- 30.7 Let Us Sum Up
- 30.8 Key Words
- 30.9 Answers/Hints to Check Your Progress Exercises
- 30.10 Exercises

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### 30.0 OBJECTIVES

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After going through this unit, you should be able to

- describe the characteristics of good estimators;
- discuss the Cramer Rao Inequality;
- explain the Asymptotic properties of the estimators; and
- estimate the value of the estimator using various methods of estimation.

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## 30.1 INTRODUCTION

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The objective of sampling is to study the features of the population based on sample observations. A carefully selected sample is expected to reveal these features, and hence we shall infer about the population from a statistical analysis of the sample. This process is known as '*statistical inference*'.

There are two types of problems. First, we may have no information at all about some characteristics of the population, especially the values of the parameters involved in the distribution, and it is required to obtain estimates of these parameters. This is the problem of '*estimation*'. Secondly, some information or hypothetical values of the parameters may be available, and it is required to test how far the hypothesis is tenable in the light of the information provided by the sample. This is the problem of '*hypothesis testing*' or 'test of significance'.

This unit will elaborate on estimation. In order to delve deep into the topic, properties of good estimators are introduced initially which are unbiasedness, consistency and efficiency. After describing these properties, the famous Cramer Rao Inequality is introduced along with the asymptotic properties of the estimators. The last sub-section of the unit deals with the methods of estimation used for ascertaining the 'best guess' of true values of the estimators.

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## 30.2 THEORY OF ESTIMATION

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Suppose, we have a random sample  $x_1, x_2, \dots, x_n$  on a variable  $x$ , whose distribution in the population involves an unknown parameter  $\theta$ . It is required to find an estimate of  $\theta$  based on sample values. The estimation is done in two different ways: (i) Point estimation, and (ii) Interval estimation.

In 'point estimation', the estimated value is given by a single numerical value, which is a function of sample observations (i.e., statistic). This function is called an 'estimator'. Since the point estimate is just a numerical figure, the value of it is less likely to coincide with the true value of the population parameter. Some degree of uncertainty is bound to be there due to sampling error. To overcome it, some intervals are constructed and it is claimed that the true value of the population parameter will lie within this interval at a confidence level of say 95% or 99%.

In 'interval estimation', an interval within which the true value of a parameter is expected to lie is computed by using two quantities based on sample values. This is known as 'confidence interval', and the quantities, which are used to specify the interval, are known as 'confidence limits'. Since our basic objective is to estimate the parameter associated with the sample conservations, so before going into further details, let us discuss about the notion of parameter space.

### 30.2.1 Parameter Space

Let us consider a random variable (r.v)  $x$  with p. d.  $f, f(x, \theta)$ . In most common application, though not always, the functional form of population distribution is assumed to be known except for the value of some unknown parameter(s)  $\theta$ , which may take any value on a set  $\Theta$ . This is expressed by writing the p. d.  $f$  in the form  $f(x, \theta), \theta \in \Theta$ . The set  $\Theta$ , which is the set of all possible values of  $\theta$ , is called the ‘parameter space’. Such a situation gives rise not to one probability distribution but a family of probability distributions which we write as  $[f(x, \theta), \theta \in \Theta]$ . For example, if  $X \sim N(\mu, \sigma^2)$ , then the parameter space  $\Theta = \{(\mu, \sigma^2): -\infty < \mu < \infty; 0 < \sigma < \infty\}$ .

For  $\sigma^2 = 1$ , the family of probability distribution is given by  $\{N(\mu, 1); \mu \in \Theta\}$ , where  $\Theta = \{\mu: -\infty < \mu < \infty\}$ . In the following discussion we shall consider a general family of distributions  $\{f(x; \theta_1, \theta_2, \dots, \theta_k): \theta_i \in \Theta, i = 1, 2, \dots, k\}$ .

Let us consider a random sample  $x_1, x_2, \dots, x_n$ , of size ‘ $n$ ’ from a population, with probability function  $f(x; \theta_1, \theta_2, \dots, \theta_k)$ , where  $\theta_1, \theta_2, \dots, \theta_k$  are the unknown population parameters. There will then always be an infinite number of functions of sample values, called statistic, which may be proposed as estimates of one or more parameters.

Evidently, the best estimates would be the one that falls nearest to the true value of the parameter to be estimated. In other words, the statistic whose distribution converges as closely as possible near to the true value of the parameter may be regarded the best estimate. Hence, the basic problem of estimation in the above case can be formulated as follows:

We wish to determine the functions of the sample observations:

$T_1 = \hat{\theta}_1(x_1, x_2, \dots, x_n), T_2 = \hat{\theta}_2(x_1, x_2, \dots, x_n), \dots, T_k = \hat{\theta}_k(x_1, x_2, \dots, x_n)$ , such that their distribution is concentrated as closely as possible near to the true value of the parameter. The estimating functions are then referred to as ‘estimator’.

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## 30.3 CHARACTERISTICS OF ESTIMATORS

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The following are some of the criteria for a good estimator: (i) Unbiasedness, (ii) Consistency (iii) Efficiency and (iv) Sufficiency. We shall now briefly explain these terms one by one.

### 30.3.1 Unbiasedness

An estimator  $(T_n)$  is an unbiased estimator of the parameter  $\theta$  if and only if  $E(T_n) = \theta$ .

The sample mean  $\bar{x}$  is unbiased estimator of the population mean,  $\mu$ , because  $E(\bar{x}) = \mu$ .

**Note:** If  $E(T_n) > \gamma(\theta)$ ,  $T_n$  is said to be positively biased and if  $E(T_n) < \gamma(\theta)$ , it is said to be negatively biased, the amount of bias  $b(\theta)$  being given by  $b(\theta) = E(T_n) - \gamma(\theta)$ ,  $\theta \in \Theta$ .

Following are some examples of unbiased and biased estimators:

- 1) If  $X$  has a binomial distribution with the parameters  $n$  and  $\theta$ , show that the sample proportion  $\frac{X}{n}$  is an unbiased estimator of  $\theta$ .

$$\text{If } E(X) = n\theta, \text{ it follows that } E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} n\theta = \theta.$$

Hence  $X/n$  is an unbiased estimator of  $\theta$ ,

- 2) If  $S^2$  is the variance of a random sample of an infinite population with the finite variances  $\sigma^2$  then  $E(S^2) = \sigma^2$ .

$$\begin{aligned} \text{Proof: By definition } E(S^2) &= E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\ &= \frac{1}{n-1} E\left[\sum_{i=1}^n \{(X_i - \mu + \mu - \bar{X})\}^2\right] \\ &= \frac{1}{n-1} E\left[\sum_{i=1}^n \{(X_i - \mu) - (\bar{X} - \mu)\}^2\right] \\ &= \frac{1}{n-1} \left[ E\left[\sum_{i=1}^n \{(X_i - \mu)\}^2\right] - nE\{(\bar{X} - \mu)\}^2 \right] \end{aligned}$$

Since,  $E\{(X_i - \mu)\}^2 = \sigma^2$  and  $E\{(\bar{X} - \mu)\}^2 = \frac{\sigma^2}{n}$  it follows that:

$$E(S^2) = \frac{1}{n-1} \left[ \sum_{i=1}^n \sigma^2 - n \frac{\sigma^2}{n} \right] = \sigma^2$$

Although  $(S^2)$  is an unbiased estimator of the variance of an infinite population, it is not an unbiased estimator of the variances of a finite population and in neither case is  $S$  an unbiased estimator of  $\sigma$ . There are difficulties associated with the concept of unbiasedness. The property may not be retained under functional transformations, i.e., if  $\phi$  is an unbiased estimator of  $\theta$ , it does not necessarily follow that  $\omega(\phi)$  is an unbiased estimator of  $\omega(\theta)$ . Another difficulty associated with the concept of unbiasedness is that unbiased estimators are not necessarily unique. Unbiasedness is a property associated with finite 'n'. A statistic  $T_n = T(x_1, x_2, \dots, x_n)$ , is said to be an unbiased estimator of  $\gamma(\theta)$  if  $E(T_n) = \gamma(\theta)$ , for all  $\theta \in \Theta$



### 30.3.2 Consistency

A desirable property of a good estimator is that its accuracy should increase as the sample size becomes larger. That is, the estimator is expected to come closer to the parameter as the size of the sample increases.

An estimator  $T_n = T(x_1, x_2, \dots, x_n)$  based on a random sample of size 'n', is said to be a consistent estimator of  $\gamma(\theta)$ ,  $\theta \in \Theta$ , if  $T_n \rightarrow \gamma(\theta)$  in probability i.e., if  $T_n P \rightarrow \gamma(\theta)$  as  $n \rightarrow \infty$ . In other words,  $T_n$  is a consistent estimator of  $\gamma(\theta)$  if for every  $\varepsilon > 0$ ,  $\eta > 0$ , there exists a positive integer  $n \geq m(\varepsilon, \eta)$  such that

$P[|T_n - \gamma(\theta)| < \varepsilon] \rightarrow 1$  as  $n \rightarrow \infty \rightarrow P[|T_n - \gamma(\theta)| < \varepsilon] > 1 - \eta; \forall n \geq m$ , where  $m$  is some very large value of  $n$ .

**Note:** If  $X_1, X_2, \dots, X_n$  is a random sample from a population with finite mean  $E(X_i) = \mu < \infty$ , then by Khinchin's weak law of large numbers (WLLN), we have

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n x_i \rightarrow E(X_i) = \mu, \text{ as } n \rightarrow \infty.$$

Hence sample mean  $(\bar{X}_n)$  is always a consistent estimator of population mean.

Obviously, consistency is a property concerning the behavior of an estimator for indefinitely large values of the sample size 'n', i.e., as  $n \rightarrow \infty$ . Nothing is regarded of its behavior for finite 'n'.

Moreover, if there exists a consistent estimator, say,  $T_n$  of  $\gamma(\theta)$ , then infinitely many such estimators can be constructed, e.g.,  $T_n' = (n-a)/(n-b)$ ,  $T_n = [1 - (a/n)] / [1 - (b/n)]$ , where  $T_n' \rightarrow T_n \rightarrow \gamma(\theta)$ , as  $n \rightarrow \infty$  and hence, for different values of  $a$  and  $b$ ,  $T_n'$  is also consistent for  $\gamma(\theta)$ .

#### Sufficient Conditions for Consistency:

Let  $\{T_n\}$  be a sequence of estimators such that for all  $\theta \in \Theta$

i)  $E_\theta(T_n) \rightarrow \gamma(\theta)$  as  $n \rightarrow \infty$

ii)  $\text{Var}_\theta(T_n) \rightarrow 0$  as  $n \rightarrow \infty$

Then  $T_n$  is a consistent estimator of  $\gamma(\theta)$ .

### 30.3.3 Efficiency

Even if we confine ourselves to unbiased estimates, there will be more than one consistent estimator of a parameter. There may be a large number of consistent estimators for  $\mu$ . Indeed, if  $T$  is consistent, so are  $T + a / \psi(n)$  and  $T \{1 + a / \psi(n)\}$ , where 'a' is any constant independent of 'n' and  $\psi(n)$  is any increasing function of 'n'. To choose among these rival estimators, some additional criteria would be needed. Thus, we may consider, together with stochastic convergence, the rate of stochastic convergence, i.e., we may demand not only that  $T$  should converge stochastically to  $\gamma(\theta)$  but also that it should do so sufficiently rapidly.

We shall confine our attention to consistent estimators that are asymptotically normally distributed. In that case, the rapidity of convergence will be indicated

by the inverse of the variance of the asymptotic distribution. Denoting the asymptotic variance by 'avar', we may say that  $T$  is the best estimator of  $\gamma(\theta)$  if it is consistent and normally distributed and if  $\text{avar}(T) \leq \text{avar}(T')$  whatever the other consistent and asymptotically normal estimator  $T'$  may be.

A consistent, asymptotically normal statistic  $T$  having this property is called 'efficient'.

### 30.3.3.1 Most Efficient Estimator

If in a case of consistent estimator for a parameter, there exists one whose sampling variance is less than that of any such estimator, it is called the most efficient estimator. Whenever such an estimator exists, it provides a criterion for measurement of efficiency of the other estimators.

**Definition:** If  $T_1$  is the most efficient estimator with variance  $V_1$  and  $T_2$  is any other estimator with variance  $V_2$ , then the efficiency  $E$  is defined as:  $E = V_1 / V_2$ . Obviously,  $E$  cannot exceed unity. If  $T_1, T_2, \dots, T_n$  are all estimators of  $\gamma(\theta)$  and  $\text{Var}(T)$  is minimum, then the efficiency of  $E_i$  of  $T_i$ , ( $i = 1, 2, \dots, n$ ) is defined as:  
 $E_i = \text{Var}(T) / \text{Var}(T_i)$ ;  $E_i \leq 1$ , ( $i = 1, 2, \dots, n$ ).

### 30.3.3.2 Minimum Variance Unbiased Estimator (MVUE)

If a statistic  $T = T(x_1, x_2, \dots, x_n)$ , based on sample of size 'n' is such that:

- i)  $T$  is unbiased for  $\gamma(\theta)$ , for all  $\theta \in \Theta$  and
- ii) it has the smallest variance among the class of all unbiased estimators of  $\gamma(\theta)$ , then  $T$  is called the minimum variance unbiased estimator (MVUE) of  $\gamma(\theta)$ . More precisely,  $T$  is MVUE of  $\gamma(\theta)$  if  $E_\theta(T) = \gamma(\theta)$  for all  $\theta \in \Theta$  and  $\text{Var}_\theta(T) \leq \text{Var}_\theta(T')$  for all  $\theta \in \Theta$  where  $T'$  is any other unbiased estimator of  $\gamma(\theta)$ .

Let us discuss some important theorems concerning MVUE.

- A MVUE is unique in the sense that if  $T_1$  and  $T_2$  are MVUE for  $\gamma(\theta)$ , then  $T_1 = T_2$ .
- Let  $T_1$  and  $T_2$  be the unbiased estimators of  $\gamma(\theta)$  with efficiencies  $e_1$  and  $e_2$  respectively and let  $\rho = \rho_\theta$  be the correlation coefficient between them, then  $(e_1 e_2) - \sqrt{(1 - e_1)(1 - e_2)} \leq \rho \leq (e_1 e_2) + \sqrt{(1 - e_1)(1 - e_2)}$
- If  $T_1$  is a MVUE of  $\gamma(\theta)$ ,  $\theta \in \Theta$  and  $T_2$  is any other unbiased estimator of  $\gamma(\theta)$  with efficiency  $e = e_\theta$ , then the correlation coefficient between  $T_1$  and  $T_2$  is given by  $\rho = \sqrt{e}$ , i.e.,  $\rho_\theta = \sqrt{e_\theta}$ , for all  $\theta \in \Theta$ .

**Example:** Show that  $\bar{X}$  is a minimum variance unbiased estimator of the mean  $\mu$  of a normal population.

$$\text{Since } f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \text{ for } -\infty < x < \infty$$